

SENSI

WEEK 8

a sensorial-comfort copilot for architectural layouts
what shipped since week 7 — the space becomes an image

△ thermal ○ visual ~ acoustic □ spatial ≈ olfactory : tactile

Six things landed

THE VISION / REPORT

scores become a prompt, then an image.

BEFORE / AFTER

scrub the glow-up >> image, scores & prompt move together.

BENCHMARKING

per-node tiers, measured; Google vs OpenAI images.

CLI ORCHESTRATOR

headless mode >> Sensi runs as a component.

CHECKPOINTS + MULTI-EDIT

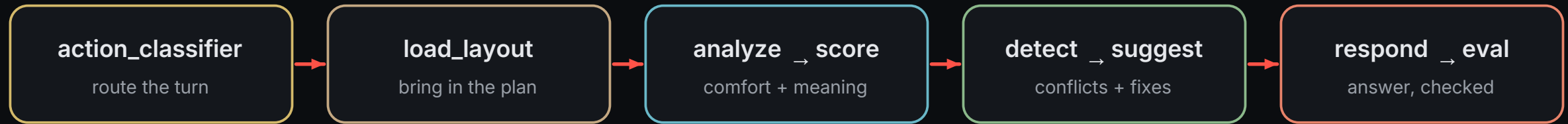
git-like commits; many edits, one re-score.

FINE-TUNINGS

orbs · linkified chat · prompts · declutter.

One classifier, many paths

A single `action_classifier` picks one of 13 actions in one LLM call — then the right chain runs.



`edit` `plan → apply → re-score → compare` `insight` `topology · biophilic · persona` `else` `follow-up · chitchat`

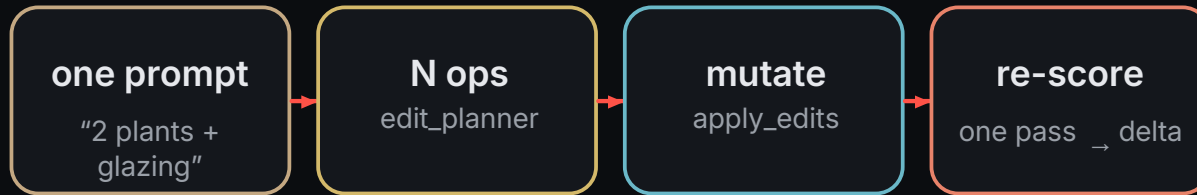
13 actions, 1 call
routing that was three LLM calls is now one.

cache-aware
no re-scoring when the layout hasn't changed.

`edit → re-score`
the agentic ripple — every change is measured.

Editing is live >> commit when you're happy

One instruction can carry several changes. `edit_planner` splits it into ops; `apply_edits` mutates them at one chokepoint >> then we re-score once.



draft → commit → restore

checkpoints are git-like — roll back anytime

[api/checkpoints.py](#)

why it matters explore freely, never lose a good state >> the Report's before/after reads the committed checkpoint.

checkpoints in the UI

UNCOMMITTED CHANGES · SINCE THE LAST CHECKPOINT

Δ 0.62→0.64 ↑

~ 0.52→0.53 ↑

name this milestone (optional)

✓ commit checkpoint

Scores become a prompt, become an image

hover on a sense to locate it in the prompt

the prompt, from the scores

the render >> how it feels

summary per persona

The screenshot displays the SENSI interface for 'the vision'. It features three main sections: 'Kitchen', 'Master Bedroom', and 'Guest Bedroom'. Each section shows a 'THE SCORES' radar chart, a 'BECOME A PROMPT' text box, and a 'BECOME AN IMAGE' image box. The 'Kitchen' section shows a 'conflicted' state with a score of 0.34. The 'Master Bedroom' shows a 'serene' state with a score of 0.66. The 'Guest Bedroom' shows a 'serene' state with a score of 0.70. Annotations with arrows point to specific elements: 'hover on a sense to locate it in the prompt' points to the 'thermal' sense in the Kitchen scores; 'the prompt, from the scores' points to the 'BECOME A PROMPT' text box in the Kitchen section; 'the render >> how it feels' points to the 'BECOME AN IMAGE' image box in the Kitchen section; and 'summary per persona' points to the 'summary per persona' text box in the Kitchen section.

Kitchen kitchen FEATURES

THE SCORES

thermal 0.34 0.30-0.34
visual 0.58 0.48-0.58
acoustic 0.19 0.18-0.19
spatial 0.87 0.80-0.87
olfactory 0.48
tactile 0.56 0.61-0.56

BECOME A PROMPT

First-person, eye-level interior photograph of a kitchen. The floor is wood. The space feels a cool, slightly cold feel with bluish daylight. Hard, reflective surfaces - bare concrete, glass and tile - that look acoustically live; open, spacious and generous in volume. Furnishings include a stone island, a ceramic counter, two potted plants. Natural perspective, 35mm lens, photorealistic, high detail, no text, no people. Shot as restrained, material-honest architectural photography.

BECOME AN IMAGE

conflicted 0.34

summary per persona

- **thermal** Emilie, who is sensitive to thermal conditions, will find the kitchen uncomfortably cold or hot.
- **acoustic** The kitchen's noise levels are extremely bothersome for Emilie, being the lowest single sense in the whole layout.
- **spatial** Emilie perceives the kitchen as very spacious and well-organized despite its other issues.
- **olfactory** Emilie, with her olfactory sensitivities, will likely find the smells in the kitchen unpleasant or overwhelming.

Master Bedroom bedroom serene 0.66

Guest Bedroom bedroom serene 0.70

THE SCORES

thermal 0.76 0.72-0.76
visual 0.63 0.60-0.63

BECOME A PROMPT

First-person, eye-level interior photograph of a bedroom. The floor is carpet. The space feels a warm, cozy feel with golden light; soft sound-absorbing textiles, rugs and drapes; open, spacious and generous in volume; fresh and well-ventilated, with a few

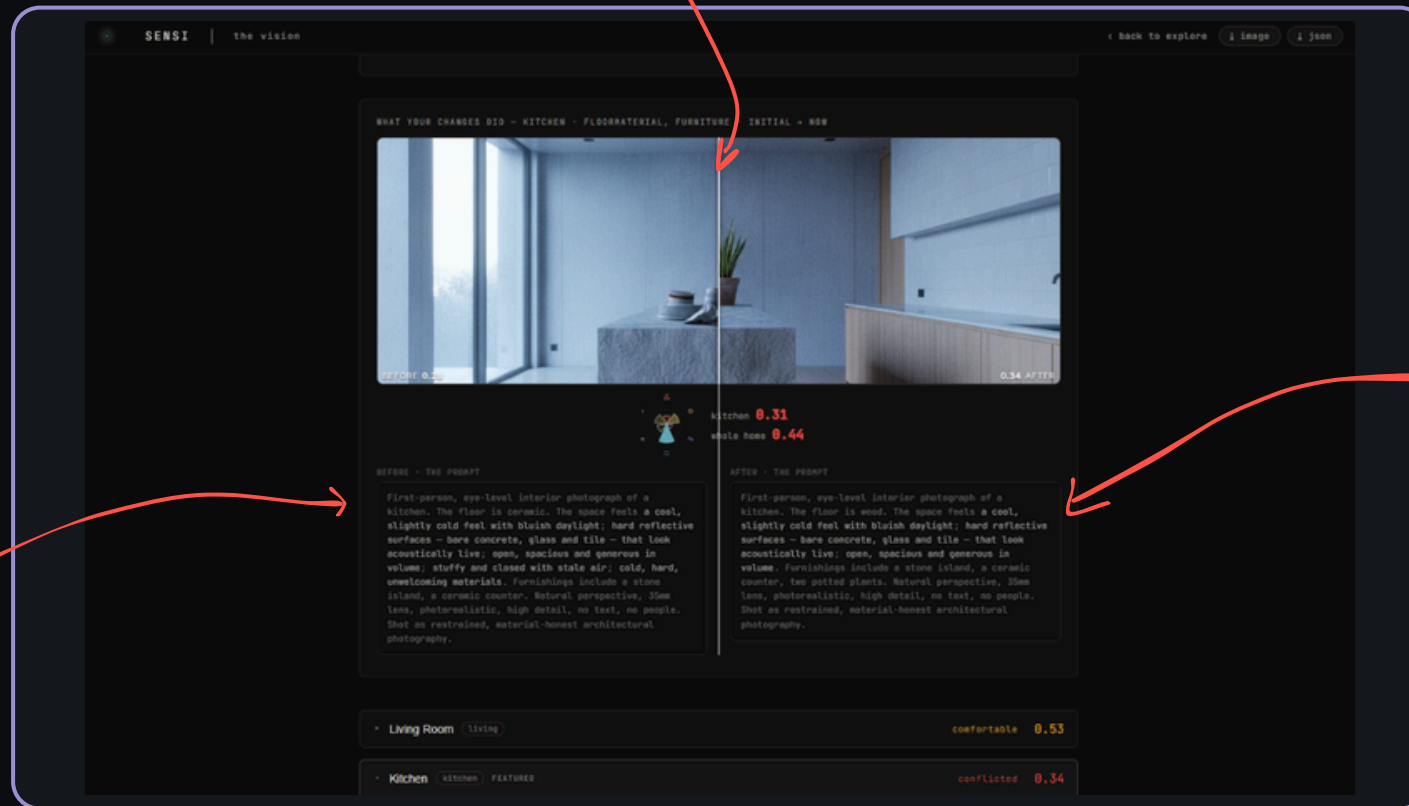
BECOME AN IMAGE

Before → after, the whole story scrubs

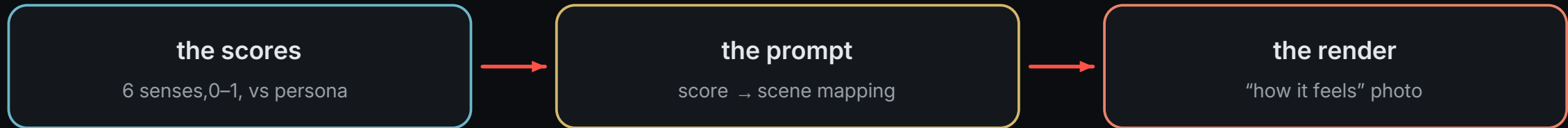
before prompt (base image)

rose + numbers morph as you drag

after prompt (text prompt +
base image)



How a room becomes an image



the rule only a room's extreme senses speak < 0.45 or > 0.70

```

First-person photo of a {room}.
The floor is {material}.
The space feels {voiced senses}.
Furnishings include {items}.
35mm, photoreal • {persona register}.
build_room_prompt()
  
```

`imaging/prompt.py`

SCORE → WORDS

~ acoustic low
"hard, reflective surfaces"

≈ olfactory high
"fresh, a few plants"

: tactile low
"cold, hard materials"

before / after the after is the room's canonical render (the card's image). The before is that image, edited by a "what changed" clause >> so the scene holds, only the changes move.

Small models for simple work

13 LLM nodes, two tiers. Measured one full session with `bench_nodes.py`:

node	tier	avg s	tok in/out	\$/call
suggestion_critic	SMART	16.1	924 / 838	\$0.0024
score_interpreter	SMART	14.3	1696 / 1024	\$0.0031
conflict_reasoner	SMART	9.3	1062 / 320	\$0.0011
respond	SMART	7.4	4820 / 243	\$0.0021
detail_respond	SMART	2.1	8017 / 42	\$0.0025
action_classifier	FAST	0.74	4105 / 175	\$0.0005
what_next	FAST	0.79	2564 / 182	\$0.0003
evaluator	FAST	0.47	1272 / 3	\$0.0001
greet	FAST	0.54	124 / 12	\$0.0000

PER-CALL AVERAGE

FAST 0.68 s · ~\$0.0002

SMART 8.48 s · ~\$0.002

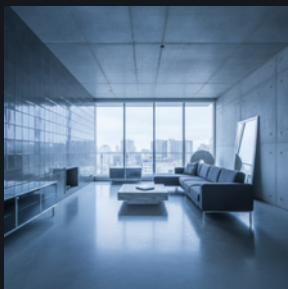
SMART ≈ 12× slower, 11× costlier.

NO-LLM NODES

analyze · detect · suggest · apply_edits — pure Python, ~0 s.

Same prompt, two engines

living > poor



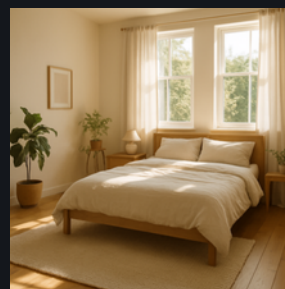
kitchen > mixed



bedroom > cosy



GOOGLE



OPENAI

THE NUMBERS

Google

6.7 s · \$0.039

OpenAI

20.5 s · \$0.042

stay on Google

~3× faster, a bit cheaper, steadier
across edits.

Both read the comfort scores >>> Google leans clean daylight, OpenAI darker and cinematic.

Many small wins

CLEANER CANVAS

the material lens is floating orbs + a ripple on every change >> not heavy fills.

CHAT TALKS TO THE PLAN

rooms, senses & scores linkify and brush the plan both ways.

SYSTEM PROMPTS REVIEWED

one shared register; exact room & sense names so the linker fires.

DECLUTTERED

removed redundancies across the report & explore views.

Where week 9 goes



tune the loop

make the evaluator earn its keep; stream the slow reasoners



attribute orbs

redesign the orbs per-attribute >> see each change, not just material



more agentic tools

add window · change wall material >> and optimize the ones we have

keep making the edges legible >>> now in pixels.